JAVA PROJECT REPORT ON

SUPERMARKET CHECKOUT AND DISCOUNT BILLING SYSTEM

Submitted by

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Under the Guidance of

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Semester: **I**

DECLARATION

I hereby declare that the project entitled "**SUPERMARKET CHECKOUT AND DISCOUNT BILLING SYSTEM"** submitted in partial fulfillment of the requirements for the course [**Course Name**] is a **record of my original work** carried out under the guidance of **[Trainer Name]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"SUPERMARKET CHECKOUT AND DISCOUNT BILLING SYSTEM"** is a **bona fide work** carried out by **[YOUR NAME] [Class/Batch]**, [**Department Name**], **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on

Guide: **Head of the Department**

(Technical Trainer, KGCAS) (Technical Head, KGCAS)

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ABSTRACT

The **Supermarket Checkout and Discount Billing System** is a Java-based application designed to automate the checkout and billing process in supermarkets. In many small supermarkets and retail shops, billing operations are performed manually, which can lead to calculation errors, delays at checkout, and inefficient discount management. This project aims to simplify the billing process by providing an automated system that calculates the total cost of purchased items, applies applicable discounts, and generates an accurate invoice.

The system allows users to scan or select products from a predefined inventory, enter the quantity of each item, and automatically calculate the total bill including applicable discounts and taxes such as GST. The application is developed using **Core Java programming concepts**, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a **console-based application or using Java Swing for a graphical user interface**.

The main objective of the system is to improve checkout efficiency, reduce human errors, and provide a faster billing experience to customers. The system generates a clear and structured bill that displays the purchased items, quantities, original prices, applied discounts, subtotal, tax, and final payable amount.

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a basic foundation for developing more advanced supermarket management systems in the future.

1. INTRODUCTION

In the retail industry, efficient checkout and billing systems play a crucial role in managing supermarket operations and ensuring customer satisfaction. Traditionally, many small supermarkets and grocery stores rely on manual billing methods, where staff calculate the total bill using handwritten notes or simple calculators. This manual process is time-consuming and can often lead to calculation errors, incorrect discount application, delays at checkout, and difficulty in maintaining proper billing records.

With the advancement of computer technology, automated billing systems have become essential for improving efficiency and accuracy in retail management. A computerized billing system helps supermarket staff quickly generate bills, apply discount schemes automatically, calculate totals, and maintain organized records of purchases and transactions.

The **Supermarket Checkout and Discount Billing System** is a Java-based application designed to automate the process of product checkout and bill generation. The system allows the user to select products from a predefined inventory, enter the quantity of items purchased, and automatically calculate the total bill including applicable discounts and taxes. The application generates a clear and structured invoice displaying the purchased items, quantities, prices, discounts, subtotal, tax, and final payable amount.

This project is developed using **Core Java programming concepts** such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a **console-based program or using Java Swing for a graphical user interface (GUI)**.

The primary goal of this system is to reduce manual effort, improve billing accuracy, apply discount schemes accurately, and provide faster checkout service to customers. This project also demonstrates how Java programming can be applied to develop simple real-world retail business applications.

1.1 Objective

The main objective of the **Supermarket Checkout and Discount Billing System** is to develop a simple and efficient billing application using Core Java that automates the process of product checkout, discount application, and bill generation in supermarkets and retail shops.

The specific objectives of this project are:

- To design and develop a **Java-based supermarket billing system**.
- To automate the **calculation of product prices, discounts, and total bill**.
- To reduce **manual errors in billing and discount calculations**.
- To provide a **user-friendly system for supermarket staff** to generate invoices quickly.
- To demonstrate the implementation of **Core Java programming concepts** such as classes, objects, methods, loops, and conditional statements.

- To generate a **clear and organized bill for customers**.

1.2 Overview

The **Supermarket Checkout and Discount Billing System** is a menu-driven application developed using the Java programming language. The system provides a digital platform where supermarket staff can scan or select products from a predefined inventory and generate bills automatically.

The system displays a list of available products along with their prices and applicable discounts. The user can select the required items and enter the quantity purchased by the customer. The program calculates the total price for each item, applies the eligible discount, and generates a final bill including applicable taxes such as GST.

The system helps simplify the checkout process by eliminating manual calculations and reducing the chances of billing errors. The application can be developed as a **console-based program or a graphical user interface using Java Swing**.

This project demonstrates the practical use of Java programming in developing real-world applications for supermarket and retail environments.

1.3 Features

The **Supermarket Checkout and Discount Billing System** includes the following main features:

- **Product Display** – Displays a list of available products with their prices and applicable discount percentages.
- **Item Selection** – Allows the user to select products from the inventory and enter the quantity purchased.
- **Discount Application** – Automatically applies eligible discount schemes based on product category or purchase amount.
- **Automatic Bill Calculation** – Automatically calculates the total amount based on selected items, quantities, and discounts.
- **Tax Calculation** – Calculates applicable taxes such as GST and adds them to the total bill.
- **Invoice Generation** – Generates a detailed bill showing item names, quantities, prices, discounts, subtotal, tax, and final amount.
- **User-Friendly Interface** – Provides a simple and easy-to-use menu-driven interface.
- **Fast Checkout Process** – Reduces the time required to process customer purchases and generate bills.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many small supermarkets and retail stores, billing is often performed manually using paper or simple calculators. This traditional method creates several challenges.

Some common problems include:

- Errors in bill calculation
- Incorrect or missed discount application
- Time-consuming checkout process
- Difficulty in managing multiple customer orders
- Lack of proper billing records
- Inefficient service during peak shopping hours

Manual billing systems also increase the workload for supermarket staff and can lead to customer dissatisfaction due to delays or incorrect billing amounts.

Therefore, there is a need for an automated billing system that can quickly calculate the total bill, apply discount schemes, manage product checkouts, and generate accurate invoices. The **Supermarket Checkout and Discount Billing System** is designed to solve these problems by providing a computerized solution that automates the billing and discount process.

2.2 Project Overview

The **Supermarket Checkout and Discount Billing System** is a Java-based application developed to simplify the checkout and billing process in supermarkets and retail shops.

The system provides a menu-driven interface where users can select products from a predefined inventory and specify the quantity of each item purchased by the customer. The system automatically calculates the total price, applies applicable discounts, and generates a final bill.

The system includes the following functionalities:

- Displaying the product inventory with prices and discount information
- Selecting products and entering purchase quantity
- Calculating item cost automatically

- Applying discount schemes based on product or purchase value
- Computing subtotal, discount amount, and tax
- Generating the final bill for the customer

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application**.

2.3 Scopes and Limitations

Scope

The scope of the **Supermarket Checkout and Discount Billing System** includes the following:

- Managing supermarket product inventory
- Processing customer product purchases at checkout
- Applying discount schemes automatically
- Calculating the total bill including taxes
- Generating a detailed invoice for the customer
- Demonstrating the practical use of Java programming concepts

The system can be used in:

- Supermarkets
- Grocery stores
- Retail shops
- Departmental stores
- Small convenience stores

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier. The **Supermarket Checkout and Discount Billing System** is designed as a menu-driven application where users can select products, enter quantities, apply discounts, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Supermarket Checkout and Discount Billing System** are explained below:

1. Product Display Module

This module is responsible for displaying the list of available products along with their prices and applicable discounts. The product list helps the user select the required items for checkout.

Example Product List:

1. Rice (5 kg) ₹250 Discount: 5%
2. Cooking Oil (1 L) ₹180 Discount: 10%
3. Wheat Flour (2 kg) ₹90 Discount: 5%
4. Sugar (1 kg) ₹50 Discount: 0%
5. Biscuits (Pack) ₹40 Discount: 15%

The product list is displayed when the program starts so that the user can easily view the available options.

2. Item Selection Module

This module allows the user to select products from the inventory and enter the quantity purchased by the customer.

Functions of this module include:

- Accepting user input for product selection
- Accepting quantity of selected products
- Storing purchase details temporarily

The system allows users to select multiple products before generating the final bill.

3. Discount Calculation Module

This module handles the application of discount schemes to the selected products.

The system supports:

- Product-level discounts (percentage based on item category)
- Cart-level discounts (e.g., 10% off on total purchases above ₹500)
- Promotional discounts on selected items

Example Calculation:

Discount Amount = Original Price × Discount %

Discounted Price = Original Price – Discount Amount

4. Bill Calculation Module

This module performs all the billing calculations.

The system calculates:

- Total price of each item after discount
- Subtotal of all selected items
- Applicable tax (GST)
- Final payable amount

The calculation is performed automatically using Java programming logic.

Example Calculation:

Item Total = Discounted Price × Quantity

GST = 5% of Subtotal

Final Bill = Subtotal + GST

5. Invoice Generation Module

This module generates the final bill for the customer.

The bill includes:

- Product name
- Quantity
- Original price
- Discount applied
- Discounted price
- Subtotal
- GST amount
- Final bill amount

Example Bill Output:

------ SUPERMARKET BILL ------

Item	Qty	Price	Disc	Total
Rice (5 kg)	2	500	5%	475
Biscuits (Pack)	3	120	15%	102

Subtotal	577
GST (5%)	29
Total	606

Thank You! Visit Again!

6. Exit Module

This module allows the user to exit the application safely after completing the checkout and billing process. When the user selects the exit option, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system. For the **Supermarket Checkout and Discount Billing System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of supermarket checkout operations.

The main problems identified in manual billing systems are:

- Manual calculations may lead to billing errors.
- Discount schemes are often applied incorrectly or missed.
- Preparing bills manually takes more time, increasing queue length.
- Difficult to handle multiple customer checkouts efficiently.
- Lack of automated records for transactions and discounts.

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies supermarket checkout and discount operations.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Product Display Module
- Item Selection Module
- Discount Calculation Module
- Bill Calculation Module
- Invoice Generation Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

- Creation of classes and methods for product and billing logic
- Implementation of menu-driven checkout logic
- Writing code for discount and bill calculation
- Handling user input and validation
- Displaying the final bill with discount breakdown

Core Java concepts such as **classes, objects, loops, conditional statements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Select product from inventory	Product displayed correctly
Enter quantity	Correct quantity accepted
Apply discount	Correct discount calculated
Calculate bill	Correct total displayed
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the selected products, quantities, and applicable discounts.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Supermarket Checkout and Discount Billing System**:

1. Start the program.
2. Display the list of available products with prices and discounts.
3. Ask the user to select a product.
4. Enter the quantity of the selected product.
5. Calculate the discount amount for the selected product.
6. Calculate the discounted price of the product.
7. Calculate the total cost of the item (Discounted Price × Quantity).
8. Add the cost to the total bill.
9. Repeat steps 3–8 for additional items if required.
10. Check if the cart total qualifies for a cart-level discount.
11. Calculate the GST or tax amount.
12. Calculate the final bill amount.
13. Display the final bill to the user.
14. End the program.

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start

↓

Display Product Inventory

↓

Select Product

↓

Enter Quantity

↓

Apply Discount

↓

Calculate Bill

↓

Display Invoice

↓

End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

Include important parts of the Java code such as:

- Main class
- Product and discount logic
- Billing calculation

Example:

```
discountAmount = price * (discountPercent / 100.0);
discountedPrice = price - discountAmount;
itemTotal = discountedPrice * quantity;
gst = subtotal * 0.05;
finalBill = subtotal + gst;
```

6. INPUT AND OUTPUT SCREENS

Include screenshots of:

- Program menu / product list
- Item selection
- Discount calculation output
- Billing output
- Final invoice

Each screenshot should be labeled.

Example:

Figure 1: Product Display Screen

Figure 2: Item Selection and Discount Screen

Figure 3: Bill Output

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Supermarket Checkout and Discount Billing System was developed to automate the checkout and billing process in supermarkets and retail shops. The system allows users to select products, enter quantities, apply discounts, and generate bills automatically.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The discount calculation module was designed to handle both product-level and cart-level discount schemes. The system successfully performs bill calculations, applies discounts accurately, and generates precise invoices.

The system was tested with different inputs to ensure that it performs the required calculations correctly. The results show that the system reduces manual effort, eliminates billing errors, and improves checkout efficiency.

7.2 Conclusion

The Supermarket Checkout and Discount Billing System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world retail business application.

The system automates the checkout and billing process, applies discount schemes accurately, reduces calculation errors, and provides faster service to customers. It simplifies supermarket billing operations by generating accurate invoices based on customer purchases.

This project helps students understand how Java programming can be used to develop practical software systems for retail environments. The project can be further enhanced by adding features such as database connectivity for inventory management, barcode scanning integration, graphical user interfaces, loyalty points tracking, and online payment systems.

8. REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.w3schools.com
- https://www.javatpoint.com